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WPC POWER DRIVER BOARD – COST-OPTIMISED MODERN VERSION – functional replacement for Williams A-12697-3 (same outline, connectors, fuses).
DPAK MOSFET solenoid/lamp drivers, TPS552872-Q1 buck-boost regulators for +5V/+12V, 4-quadrant insulated triacs, SMD passives, 5x20 mm cartridges plus SMT TA7 F112.
Sheets: Power_Supply, CPU_Interface, Sol_HighPower (1-8), Sol_LowPower (9-16), Sol_Flashers (17-24), Sol_GeneralPurpose (25-28),
Lamp_Columns, Lamp_Rows, GI_Triacs. Every part carries Manufacturer / MPN / JLCPCB code and library / sourcing fields fields (see BOM). See README.md.

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CPU_Interface

File: cpu_interface.kicad_sch

Sol_HighPower

File: sol_highpower.kicad_sch

Sol_LowPower

File: sol_lowpower.kicad_sch

Sol_Flashers

File: sol_flashers.kicad_sch

Sol_GeneralPurpose

File: sol_gp.kicad_sch

Lamp_Columns

File: lamp_columns.kicad_sch

Lamp_Rows

File: lamp_rows.kicad_sch

GI_Triacs

File: gi_triads.kicad_sch

Modern re-implementation of the Williams A-12697-3 (same form factor)

Sheet: /
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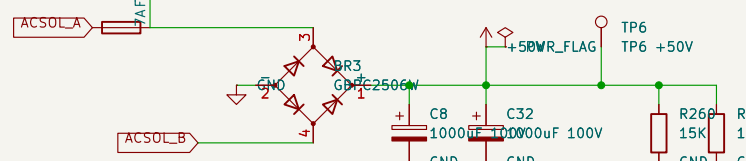
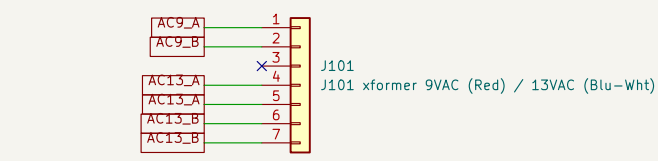
Title: WPC Power Driver Board – cost-optimised modern re-implementation

Size: A3Date: 2026-09-11Rev: JLC-100

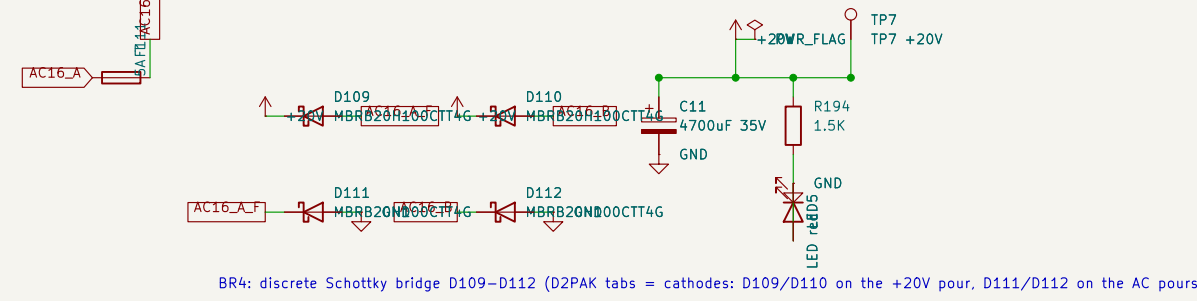
KiCad E.D.A. 10.0.6Id: 1/10

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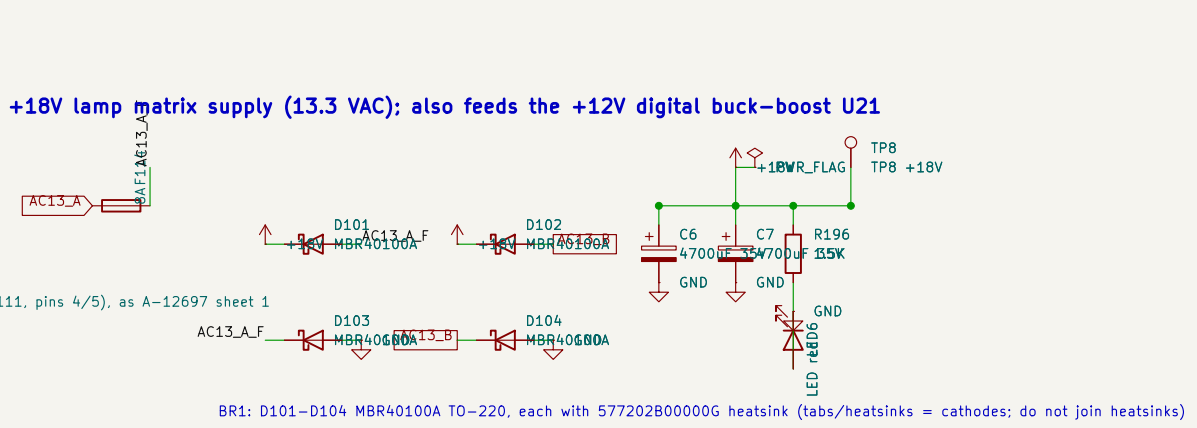
Power supplies (windings per manual/pinwiki: 9 VAC red -> +5V, 12.3 VAC blue-wht -> +18V, 16 VAC wht-red -> +20V, 51 VAC blk-gel -> +50V* (-70 V DC), 9.8 VAC wht-grn -> +12V power). Each secondary is fused (F111-F114, F116, F112 is SHT 18V, others 5x20 mm) before its bridge: 68P/2505W black (B03, 62236C basket heatsink), the +18V (D101-D104), B81, D101-D104 use MBR40100A TO-220AB with individual 57702B00000G heatinks; D105-D116 retain MBB820H00CT14G 0.294K diodes on copper pours, +50V (4 x 1000uF/100V, bleeder R299|R260) feeds the coils through F103/F104/F105 (J107-1 also feeds the 8-driver board), +20V (BRV,C11) feeds the flashlamps only (coin-door interlock kills +20V/+50V), +18V (BR1/C6/C7) feeds the lamp matrix AND the +12V digital buck-boost U21 (0.75 A load; 1.52 A nominal current limit, F115 3/4 A -> J114 to CPU J210 / Fliptronic J904 / 8-driver, as the original 7812 did), +12V power (BR5/C30+C63 9400uF, unregulated, F116) -> J116/J117/J118 pin 2, (coin door, DMD controller, the playfield optics and the SITNG gun motor), 50V BR2 -> C5 -> TP55/TP7, Q1 U20 (5.1 V, 3 A load; 3.85 A nominal current limit) -> J114/J116/J117/J118, LED1 +12V digital, LED2 +12V after F115, LED4 +5V, LED5 +20V, LED6 +18V, LED7 +12V power, TP1 12V power, TP2 5V, TP3 12V digital, TP5 GND, TP6 50V, TP7 20V, TP8 18V, TP9 5V raw, +50V solenoid supply (51 VAC winding -> -70 V DC, 78 V peak unloaded)



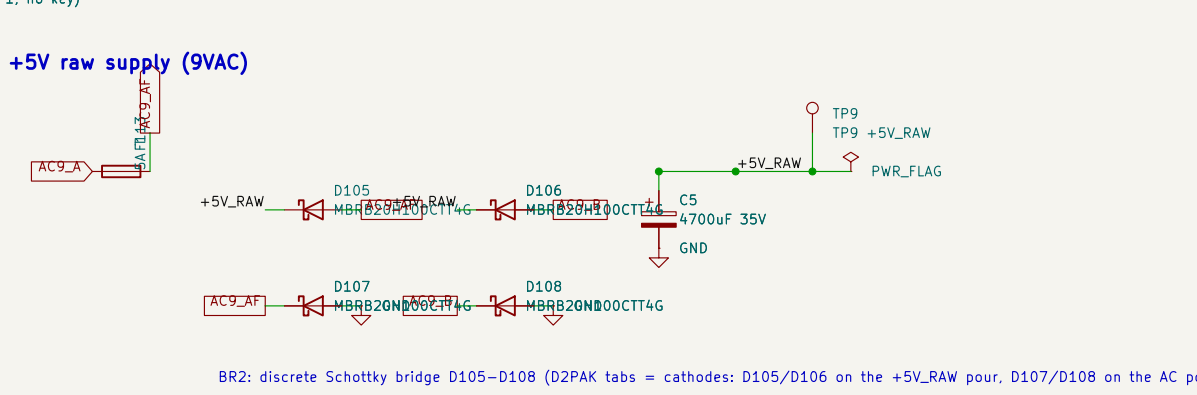
+20V flashlamp supply



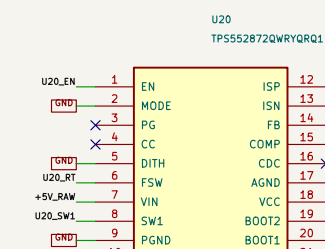
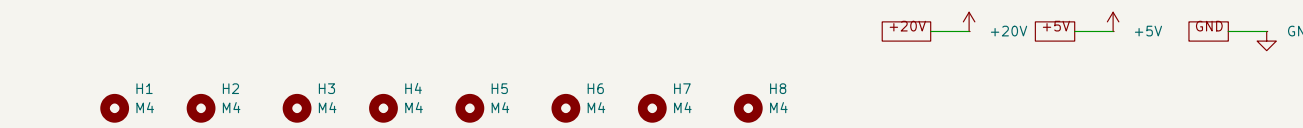
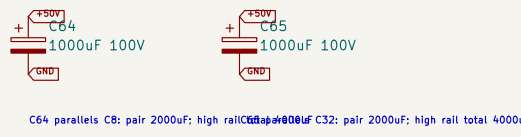
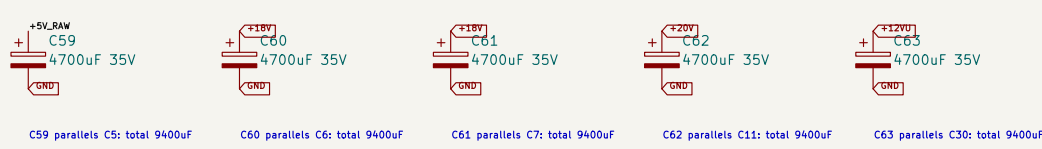
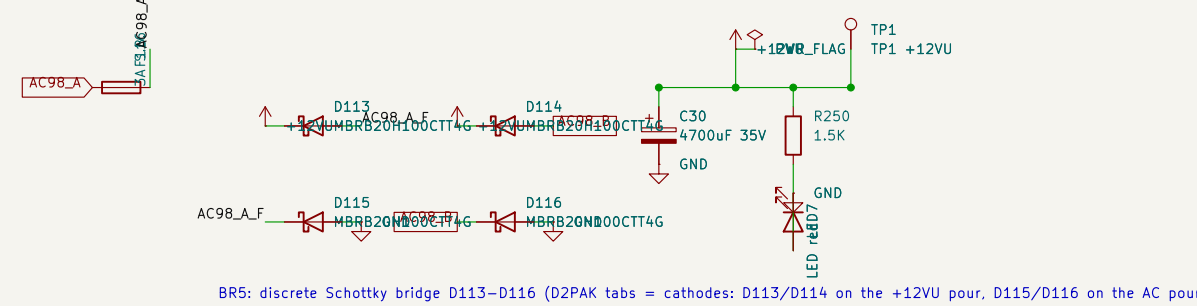
+18V lamp matrix supply (13.3 VAC); also feeds the +12V digital buck-boost U21



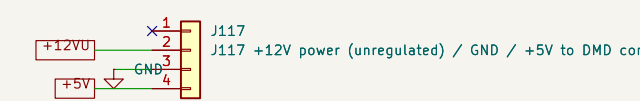
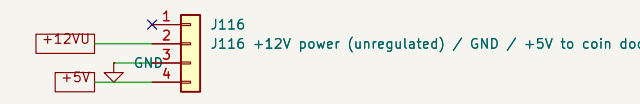
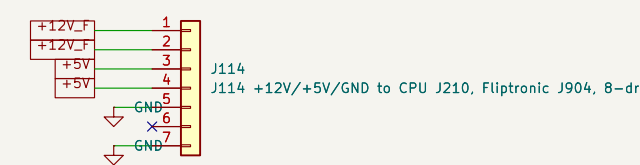
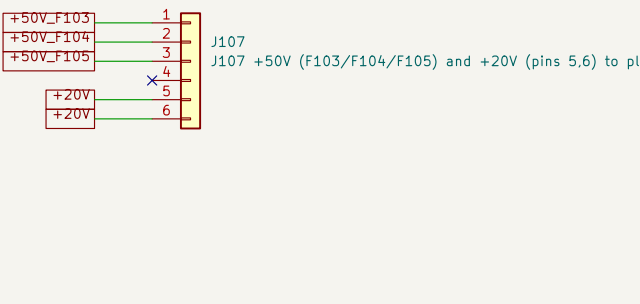
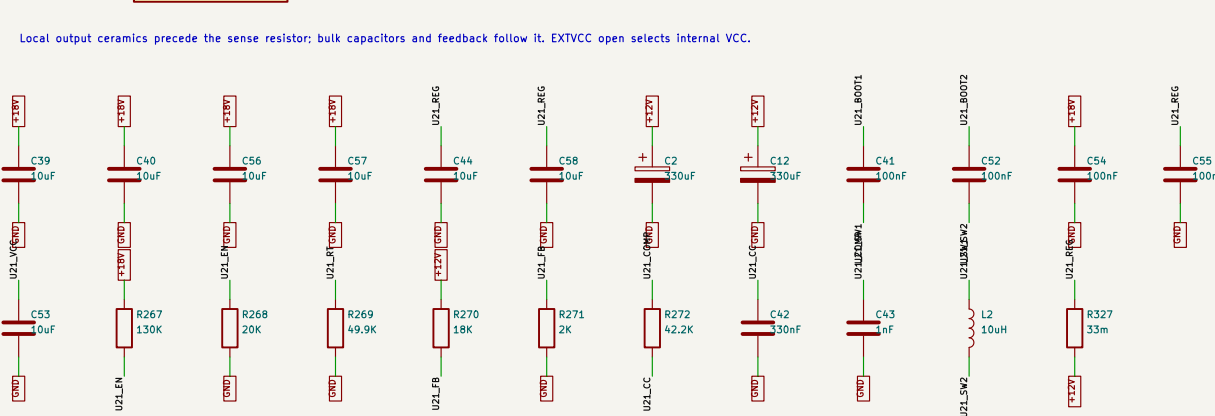
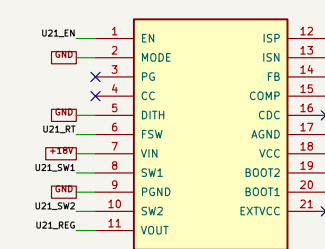
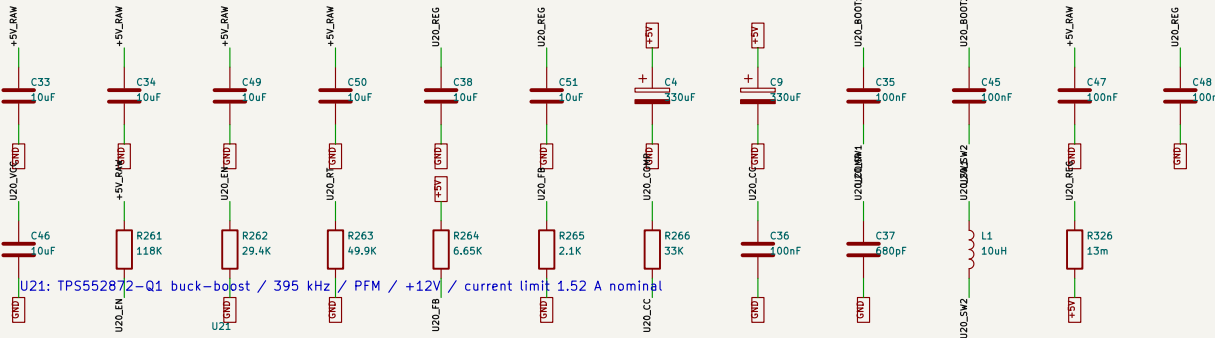
+5V raw supply (WAC)



+12V POWER (unregulated, 9.8VAC); motors, optics, coin door, DMD - J116/J117/J118



Local signal grounds provide the power isolation, both capacitors and feedback follow G, 0VDC open circuit thermal VCC.

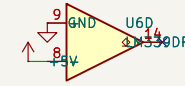
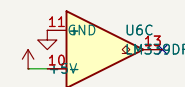
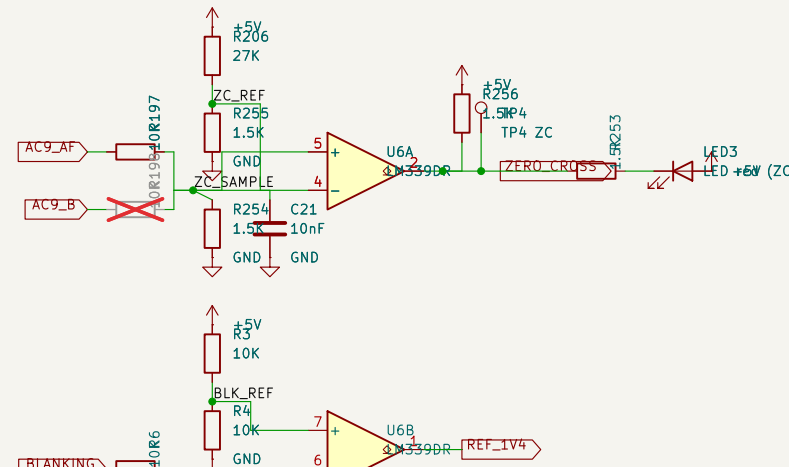
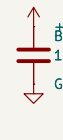
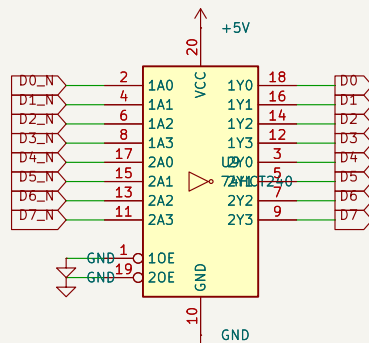
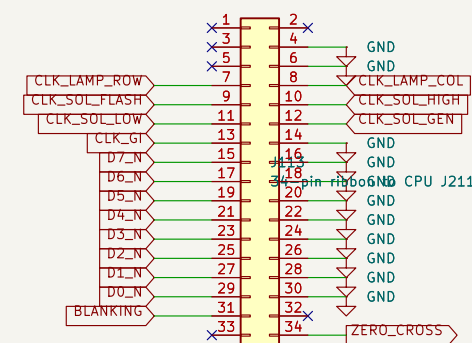


CPU interface, pin-out from the WPC schematic manual 16-9834.2. J113: 7 /LMP ROW, 8 /LMP COL, 9 /SOL 2 (sol 17-24), 10 /SOL 4 (sol 1-8), 11 /SOL 3 (sol 9-16), 12 /SOL 1 (sol 25-28), 13 /TRIAC, 15,29 odd = /D7../D0, 31 BLANKING (HIGH = every latch tri-stated, open-collector from the CPU, pulled up here).

34 ZERO CROSS to the CPU; 1/3/5/32/33 (tripper-opto sense of the -1 relay board) and 2 are N/C on the A-12697-3; other even pins = ground. The CPU board drives the data through a 74LS240, so the ribbon data is ACTIVE LOW; U9 (74HCT240) inverts it and every latch then takes the

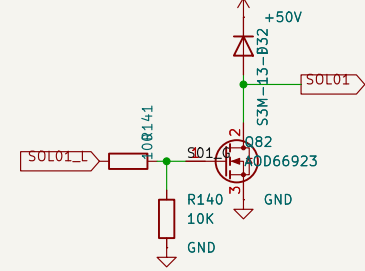
true bus (register bit = 1 -> output ON, as PinMAME / FreeWPC write it). BLANKING tri-states every 74HCT574 (gate pulled down to 0V, as bus is high-Z, and 4.7k / 15k Basic resistors; unused fifth element on the lamp-row 1.4 V reference low. Zero-cross: the F113-fused leg of the 9 VAC winding is sampled through R197 10k / R254 1.5K / C21 (R198 DNP)

the temp. low 1.4 V reference low, zero crossing the 1115 1050 leg of the 3 VAC winding's sampled through R17, 10K, 125 15K / 521 (R24) into the + input of U6A against the 0.26 V reference on - -> ZERO CROSS is HIGH while that leg is positive; a 60 Hz square wave (-7.7 ms high with one edge per mains zero crossing - the waveform of the A-12697 (U6A pin 7 +, D38 sample; a real TP4 measure of the high pulses).

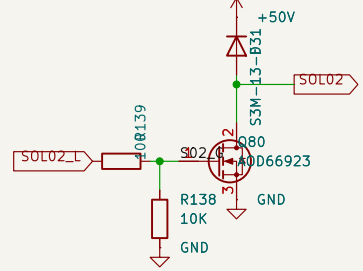


High power solenoid drivers (sol. 1–8): A0D66923 DPAK (100 V, 15 mOhm max at 4.5 V/25 C) driven directly by latch U5 (74HCT574, output HIGH = ON, outputs tri–stated by BLANKING →> gate pull–downs hold every driver OFF). SMC 3 A tie–back diodes return to +50V (the “+50V” rail of a WPC machine is ~70 V DC; 51 VAC winding, 72–78 V peak). U5 is clocked by /SOL 4 (J113–10). Output connectors J130 (playfield, key pin 3), J132 (backbox, sol 1–4, key 4) and J131 (cabinet, sol 5–8, key 2) as on the A–12697. Supply +50V via F105 on J107–3.

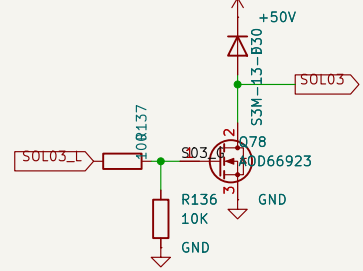
Sol 1 (high power)



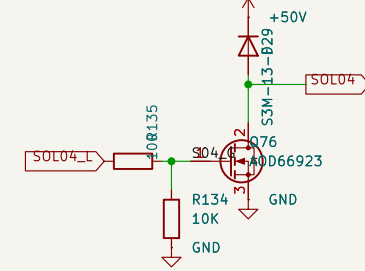
Sol 2 (high power)



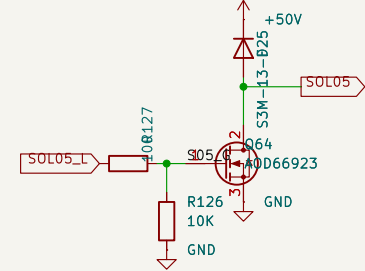
Sol 3 (high power)



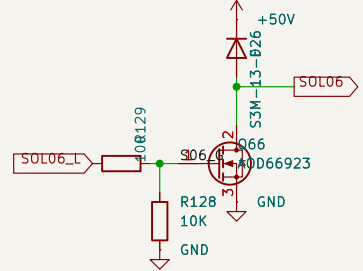
Sol 4 (high power)



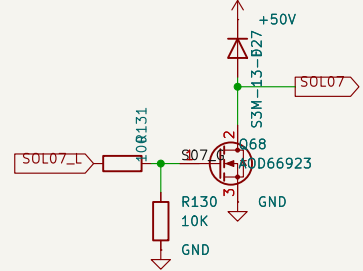
Sol 5 (high power)



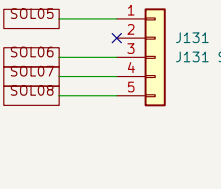
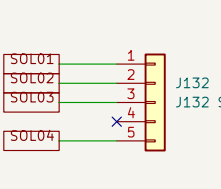
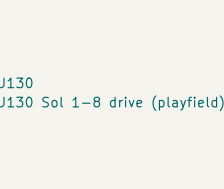
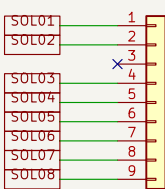
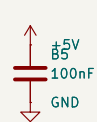
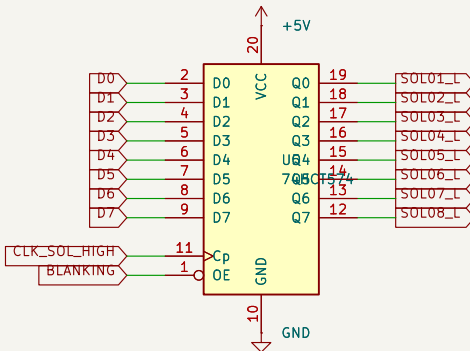
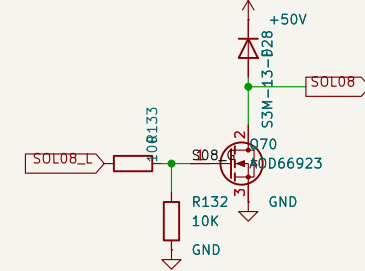
Sol 6 (high power)



Sol 7 (high power)

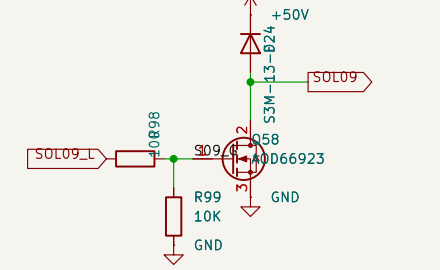


Sol 8 (high power)

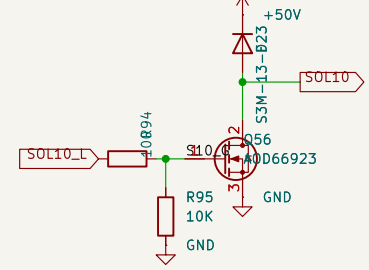


Low power solenoid drivers (sol. 9–16): AOD66923 logic–level DPAK MOSFETs (15 mOhm max at4.5V25C), latch U4 output HIGH = ON, clocked by /SOL 3 (J113–11). Tie–back diodes S3M (SMC, 3 A) to +50V (~70 V DC in the machine); coils in this group can be PWM–held by the firmware, the diode then carries the freewheel current. Output connectors J127 (playfield, key pin 2), J129 (backbox, sol 9–12, key 3) and J128 (cabinet, sol 13–16, key 4) as on the A–12697. Supply +50V via F104 on J107–2.

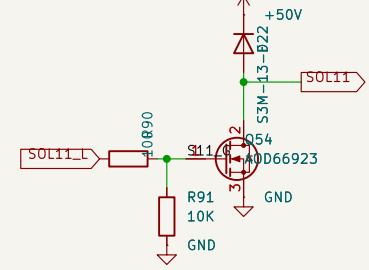
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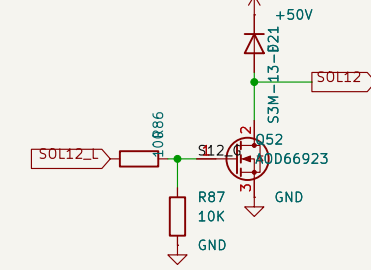
Sol 10 (low power)



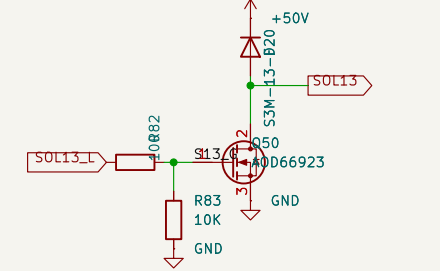
Sol 11 (low power)



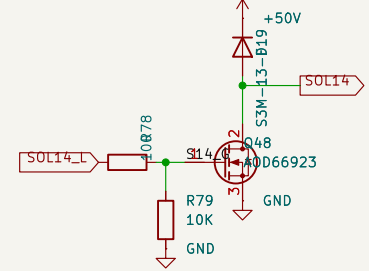
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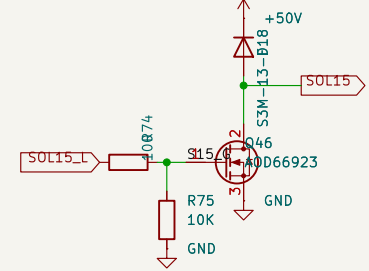
Sol 13 (low power)



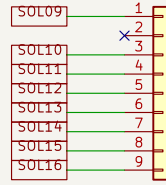
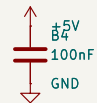
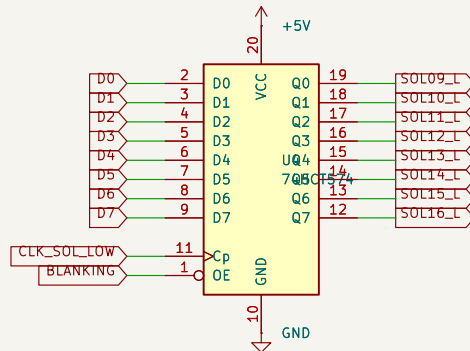
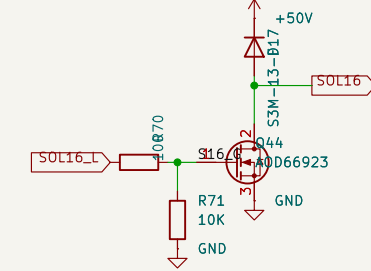
Sol 14 (low power)



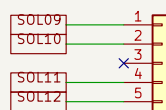
Sol 15 (low power)



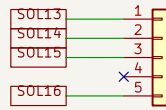
Sol 16 (low power)



J127 Sol 9–16 drive (playfield)



J129 Sol 9–12 drive (backbox, parallel to J127; N/C on STING)



J128 Sol 13–16 drive (cabinet, parallel to J127; N/C on STING)

Modern re–implementation of the Williams A–12697–3 (same form factor)

Sheet: /sol/LowPower/
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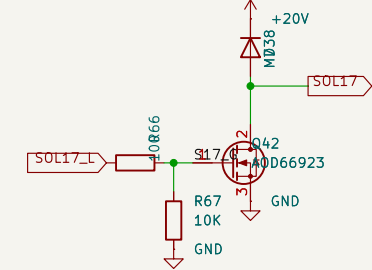
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Size: A1 Date: 2026–09–11
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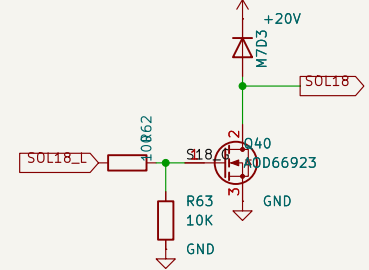
Rev: JLC–100
10/ 5/10

Flashlamp drivers (sol. 17-24): A0D66923 DPAK, latch U3 output HIGH = lamp ON, clocked by /SOL 2 (J113-9). Sol 17-20 keep an on-board 51M tie-back to +20V (the A-12697 has none there); the sol 21-24 tie-back cathodes are brought out to J126-10..13 exactly like the A-12697 D9-D12, so the harness chooses the clamp rail. On STING sol. 17/18 drive the 12 V gun motors (motor high side on the +12V power rail of J118) through this group. Outputs on J126 (playfield, key pin 9) and J125 (backbox, key pin 4). Flasher supply +20V on J107-5/6 / J106-5.

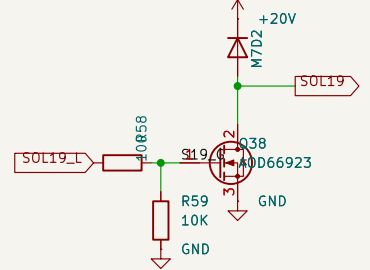
Sol 17 (flasher)



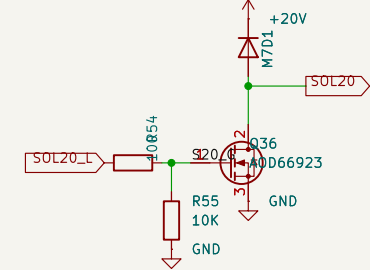
Sol 18 (flasher)



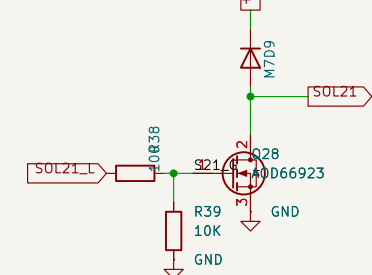
Sol 19 (flasher)



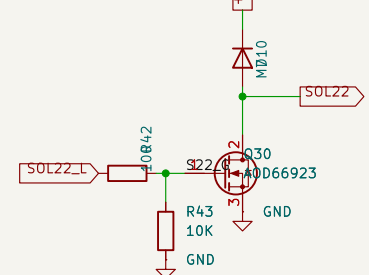
Sol 20 (flasher)



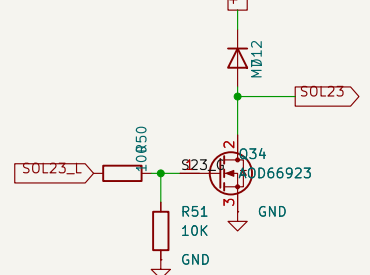
Sol 21 (flasher)



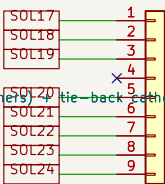
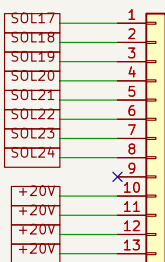
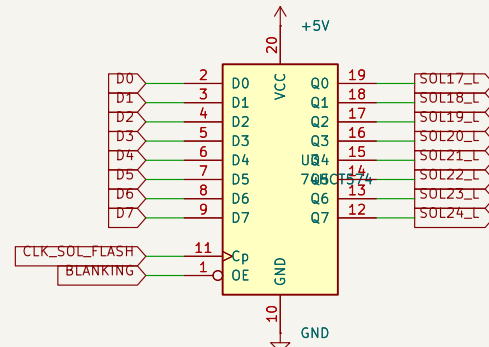
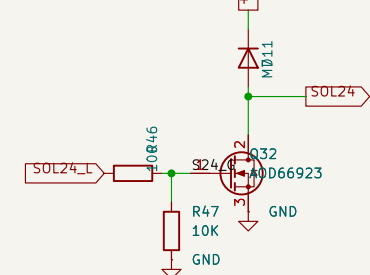
Sol 22 (flasher)



Sol 23 (flasher)

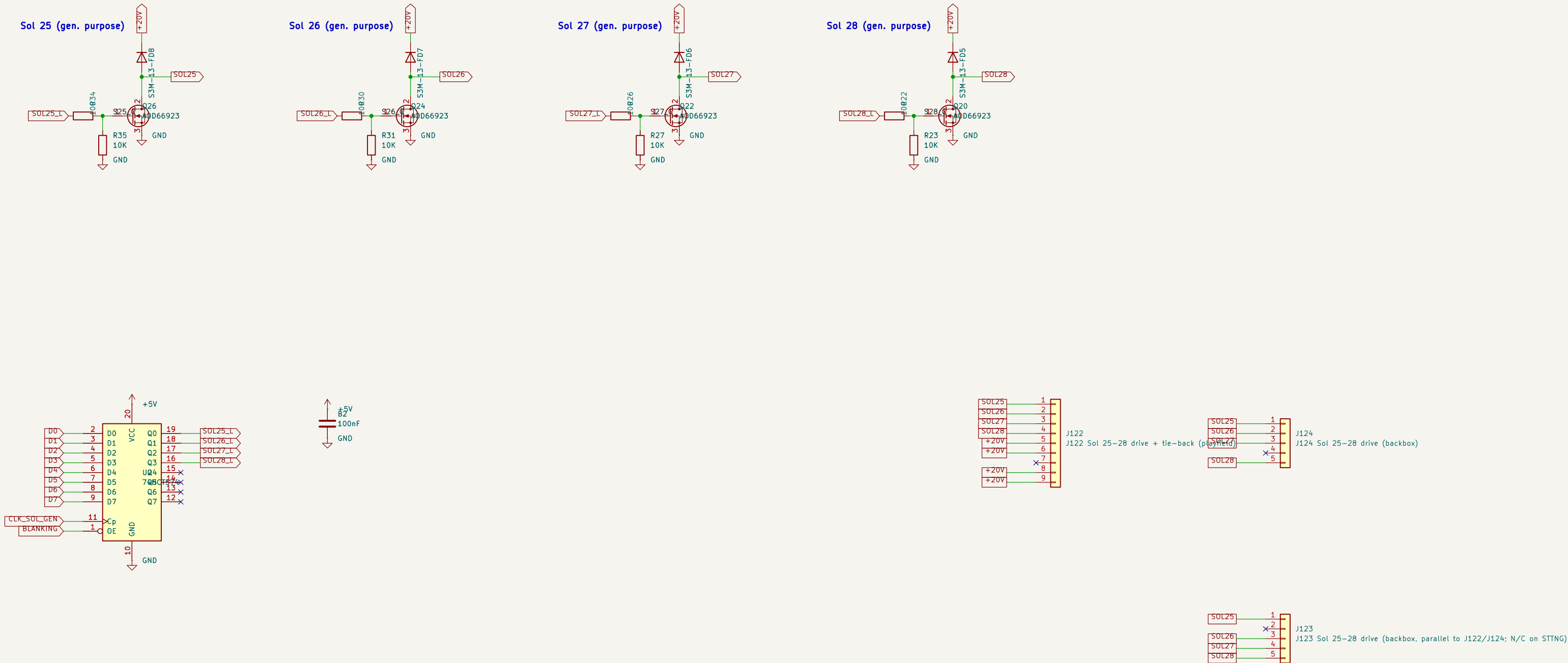


Sol 24 (flasher)

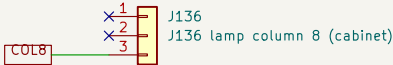
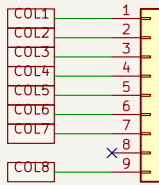
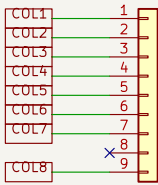
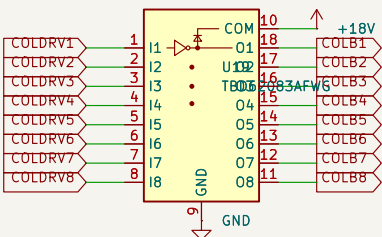
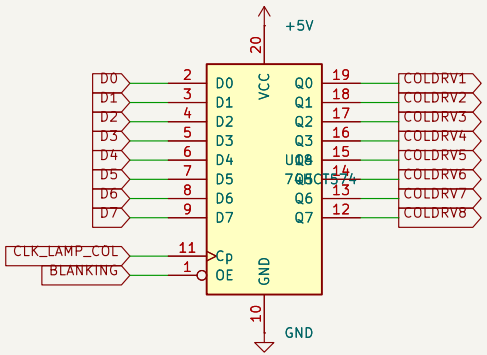
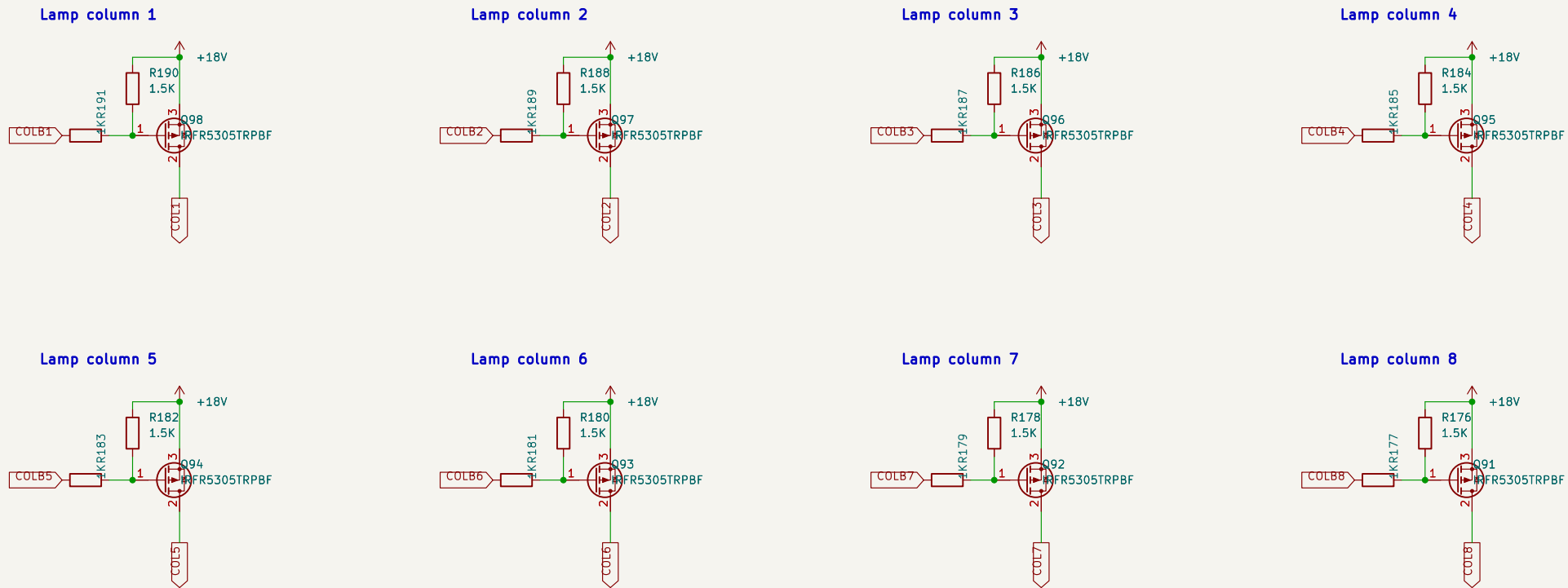


J126 Sol 17-24 drive (playfield flasher) J125 Sol 17-24 drive (backbox flasher)

General purpose drivers (sol. 25–28): AOD66923 DPAK, latch U2 output HIGH = ON, clocked by /SOL 1 (J113–12). S3M tie-back diode cathodes are brought out to J122 pins 5,6,8,9 so the harness can tie them to the supply actually used (+50V coil or +20V flasher). Outputs J122 (playfield, key pin 7), J123 (backbox, key pin 2) and J124 (cabinet, key pin 4) as on the A–12697. U2 bits 4–7 unused.



Lamp matrix column drivers. Column strobe register 0x3FE5: bit(k-1)=1 selects column k, clocked by /LMP COL (J113-8). The TBD62083A inputs have an internal pull-down (Toshiba usage note), so a tri-stated U18 (BLANKING) leaves every column OFF – the OEM 74LS374/ULN2803 pair behaves the same way. U18 latches the data bus; the TBD62083A DMOS sink pulls the P-MOSFET gate down through 1 k against the 1.5 k gate-source resistor (Vgs = -0.6 x V+18 = -8..-12 V) -> column high (+18V) with -65 mOhm (IRFR5305 DPAK; TIP107 dropped -1.2 V). Column k = Q(99-k): col1=Q98 ... col8=Q91. Outputs J137/J138 pins 1-7,9 (key pin 8).



Modern re-implementation of the Williams A-12697-3 (same form factor)

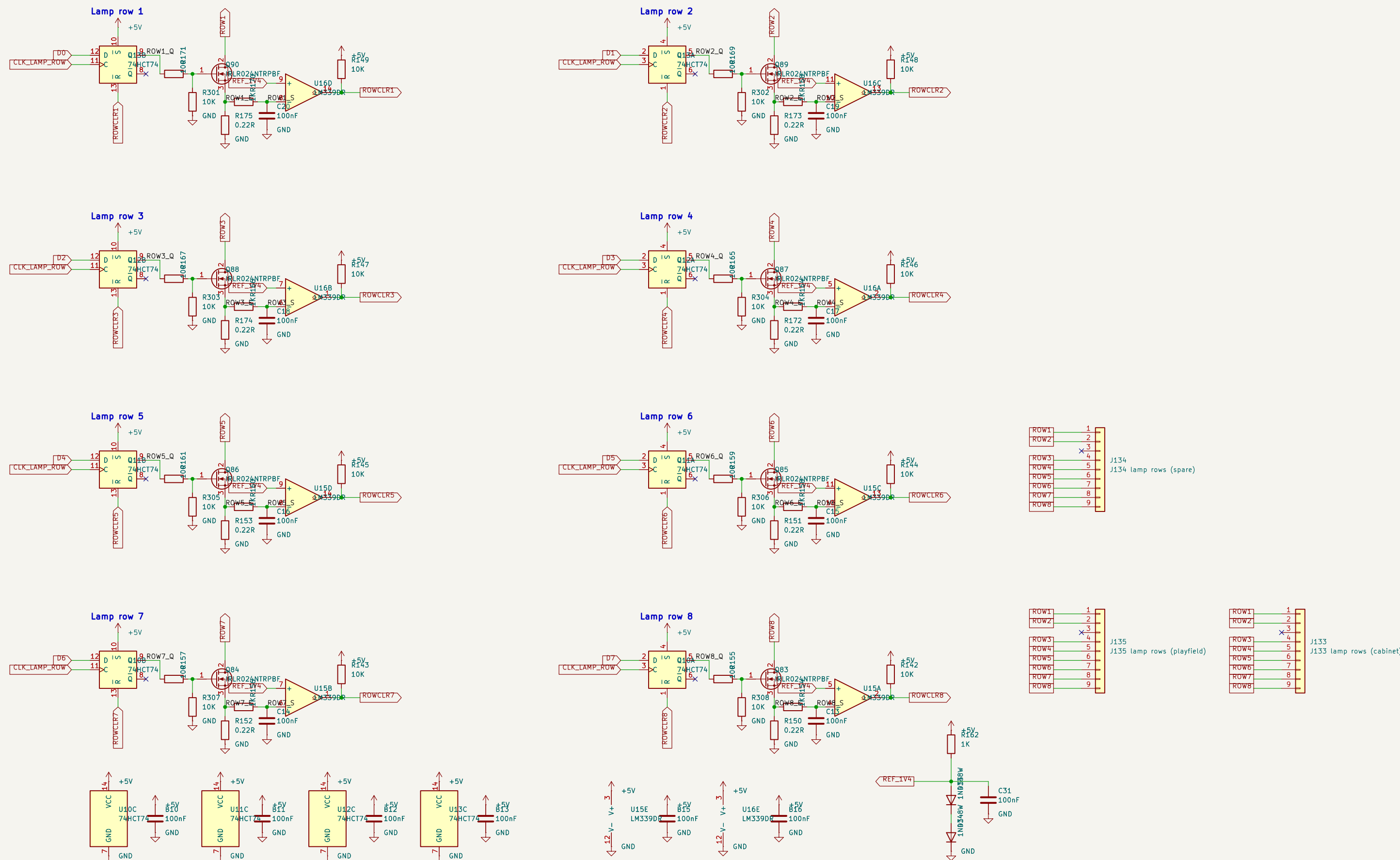
Sheet: /Lamp_Columns/
File: lamp_columns.kicad_sch

Title: Lamp matrix columns (IRFR5305 high side, +18V)

Size: A2
KiCad E.D.A. 10.0.6

Rev: JLC-100
Id: 8/10

Ramp matrix row drivers. Row register 03FEF: bit(k-4) is turns row k on, clocked by /LMP_ROW_1J133-7. The 74HC714 latches the data bit (true after U9); Q high -> IRLR024N on (p = ground, -65-80 mVrms DPAK). Overcurrent: source sense 0.22 R -> 1k/100n -> LM339; above the 1.4 V reference (6 A) the open-collector outputs pin1-C1R low and the row is shut off until the next strobe. BLANKING pins REF_1V4 low (CPU sheet) so all rows are forced off.
 Row k = 0: row1=Q90 ... row8=Q83. Outputs 1J33/1J35 (pins 1.2,4-9; key 3).



General illumination: each 6.3VAC string is fused (F106–F110) on the hot side and switched by a triac in the RETURN path; MT1 of the five triacs form the G_LRET bus that goes back to the transformer on J115–7/8/10/11/12 and is referenced to logic ground by W1 (OEM: J115–1 GND REF), so the 5 x 5 A return current stays off the ground plane. Latch U1 (clocked by TRIAC, J113–13) output HIGH = string ON. The gate is fed by an NPN emitter follower from +5V through 27 R (–80 mA, positive gate current in quadrants I and IV → 4–quadrant “C”-grade triac; snubberless BW/CW types cannot fire in Q IV). 10 k base pull–downs keep the strings OFF while the latch is tri–stated by BLANKING. Bits 5–7 (T5–T7) go to J111 (GPIO, CPU sheet); bit 7 also drives the optional flipper relay (DNP).

G.I. string #1 (Wht–Brn / Brn)

G.I. string #2 (Wht–Org / Org)

G.I. string #3 (Wht–Yel / Yel)

G.I. string #4 (Wht–Grn / Grn)

G.I. string #5 (Wht–Vio / Vio)

